Chemical Identity Lost in Regulation: A Study of Semantic Interoperability in European Chemical Substance Data

Abstract. Chemical substance identity is inconsistently represented across European regulatory datasets, creating fundamental barriers to semantic interoperability. While structure-based identifiers such as InChI and InChIKey unambiguously define chemical entities, regulatory systems rely on heterogeneous identifiers, including CAS numbers, EC numbers, textual names, and group descriptions. These identifiers often refer to mixtures, substance groups, or analytical parameters rather than uniquely defined molecular structures.

In this study, we analyse interoperability challenges across 18 European regulatory datasets from ECHA and related sources. Where possible, we map entries to InChIKeys, classify records by structural linkability, assess the consistency of CAS number mappings, evaluate embedding-based semantic matching, estimate the effort required for manual reconciliation, and implement a knowledge-graph-based integration approach.

We find that only 62.5% of entries can be directly linked to a defined chemical structure, leaving 37.5% inherently non-structure-based. CAS numbers exhibit both one-to-many and many-to-one mappings with molecular structures, undermining their role as unique identifiers. Embedding-based approaches fail to recover chemical identity for non-structure substances, and the best model achieves a ChemOnt Hit@1 of only 5.75%. The combinatorial complexity of cross-dataset identifier mappings makes retrospective harmonisation infeasible at scale.

These findings demonstrate that semantic interoperability in chemical regulation cannot be achieved through post hoc technical solutions alone. Instead, it must be addressed at the level of data modelling and legislation, by adopting structure-based identifiers, formal ontologies, and machine-readable representations embedded directly in regulatory data standards. To support further research and reuse, we provide scripts for constructing an integrated knowledge graph spanning the analysed regulatory sources.

Keywords. semantic interoperability, regulatory interoperability, Interoperable Europe, chemical substance identification

1. Introduction

Chemical substances are central entities in environmental, health, and product regulation. Across European legislation, substances are used to define obligations, assess risks, and communicate hazards in frameworks such as the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) [1], the Classification, Labelling and Packaging of chemicals (CLP) [2], and sector-specific regulations including the EU Pesticides Regulation [3] and the Water Framework Directive [4]. Effective governance of these substances increasingly depends on the ability to integrate and reuse data across regulatory domains.

In chemistry and cheminformatics, chemical identity is well-defined and operationalised through algorithmically defined, structure-based, representations. Molecular structures can be uniquely encoded using standards such as SMILES notations [5], IUPAC names [6], the International Chemical Identifier (InChI) [7] and its hashed representation, the InChIKey, enabling unambiguous identification, comparison, and linking of substances across databases [8]. Large-scale infrastructures such as PubChem [9] and ChEBI [10] build on these identifiers to provide machine-readable, ontology-based knowledge graphs that support interoperability and automated reasoning.

In contrast, European regulatory data systems adopt a fundamentally different approach. Substances are typically identified using heterogeneous and often ambiguous identifiers, including Chemical Abstracts Service (CAS) registry numbers [11,12], European Community numbers (EC-numbers) [13], textual names, and informal group descriptions. Moreover, the concept of a *substance* in regulation is not uniform: it may refer to a single molecule, a mixture of specific molecules, a mixture of unknown or variable composition (UVCB) [14], a group of related compounds (e.g., PFAS), or even an analytical parameter defined as a function over multiple substances. For a number of *substances*, the true identity, Substance Identity Profile (SIP), analytical data about the composition of *UVCB substances*, detailed usage information and full toxicological information is not disclosed to the public in accordance to REACH Artikel 118 [15] and only a competent authority has acces to complete REACH-dossiers. The general public and regional authorities have to rely on public ECHA info, CLP-classifications and safety data sheets (SDS) [16,17] which are not machine readable. Figure 1 illustrates this contrast: the right panel shows the chemistry view, in which a substance corresponds to a molecule unambiguously identified by structure-derived identifiers such as InChI or InChIKey; the left panel shows the regulatory view, where the same term encompasses heterogeneous entities of fundamentally different semantic types.

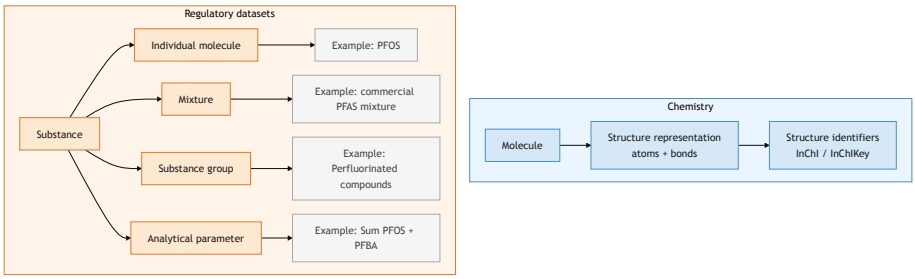

Figure 1. Conceptual model of the semantic mismatch between chemistry (right) and regulatory datasets (left). In chemistry, a substance corresponds to a molecule identified using structure-derived identifiers such as InChI or InChIKey. In regulatory datasets, the concept of a "substance" encompasses heterogeneous entities (individual molecules, mixtures, substance groups, and analytical sum parameters) that correspond to fundamentally different semantic types, complicating interoperability and automated reasoning. This mismatch is quantified empirically in Section 4.

As a result, integrating substance information across regulatory frameworks becomes a complex and largely manual task, hindering policy enforcement for environmental administrations such as the Flemish Department of Environment and Spatial Development [18], and increasing uncertainty for industrial players in environmental man-

agement, investment, and innovation. Figure 2 illustrates this concretely using PFAS as an example: individual molecules such as PFOS, PFBA, and PFBS are grouped into a regulatory substance group without explicit set-membership semantics, while an analytical parameter such as *Sum PFOS + PFBA* is a derived entity defined as a function over individual members. Neither relationship is formally specified, leading to ambiguity and combinatorial complexity during cross-dataset harmonisation.

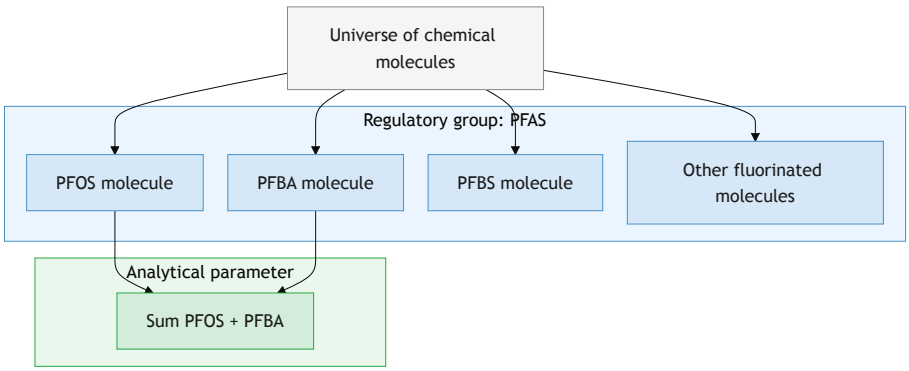

Figure 2. Set-theoretic view of a regulatory substance group (PFAS, upper subgraph) and a derived analytical parameter (lower subgraph). Individual molecules are grouped without explicit set-membership semantics, and the sum parameter aggregates contributions from named members. The absence of formal set-based definitions leads to unclear inter-entity relationships and combinatorial complexity during harmonisation.

This paper investigates the extent and consequences of this interoperability problem using a comprehensive analysis of 18 regulatory datasets from the European Chemicals Agency (ECHA) and related sources. We focus on three main research questions:

RQ1 To what extent can substances in European regulatory datasets be linked to a precise, unique chemical structure?

RQ2 How consistent and reliable are commonly used identifiers such as CAS numbers for cross-dataset linking?

RQ3 Can semantic or embedding-based methods compensate for the absence of structure-based chemical identifiers?

Based on this analysis, we make the following contributions:

C1 We quantify the heterogeneity of regulatory substance definitions and introduce a taxonomy of linkability reflecting their degree of chemical specificity.

C2 We demonstrate the limitations of CAS-based identification and show that a substantial fraction of regulatory substances cannot be resolved to a uniquely defined molecular structure.

C3 We evaluate semantic embedding approaches and show that they fail to recover chemical identity for non-structure substances.

C4 We estimate the scale of the manual effort required for retrospective harmonisation, demonstrating its infeasibility.

C5 We show that structure-based identifiers combined with ontology-driven approaches enable scalable integration through knowledge graphs.

The remainder of this paper is structured as follows. Section 3 describes the data sources and analytical methods. Section 4 presents the empirical findings. Section 5 discusses the implications for regulatory data infrastructures and semantic interoperability. Section 6 concludes with recommendations for future work.

2. Related Work

Interoperability of scientific data has been extensively studied in the context of FAIR principles [19] and knowledge graph infrastructures. In the life sciences, large-scale resources such as PubChem [9] and the Chemical Entities of Biological Interest (ChEBI) ontology [10] demonstrate how chemical entities can be represented using structure-based identifiers, ontology-driven classifications, and machine-readable knowledge graphs, enabling consistent integration, querying, and reuse of chemical information across scientific, environmental, and regulatory domains. Rule-based chemical classification is further exemplified by ClassyFire [20], which establishes a structure-based taxonomy (ChemOnt) in which each class is characterised by precise and unambiguous structural rules.

Within the Semantic Web community, numerous approaches have been proposed for data integration based on linked data principles, ontology alignment, and entity resolution. Techniques such as identifier mapping, schema matching, and, more recently, embedding-based similarity methods [21] have been applied to address heterogeneity across datasets. However, these approaches typically assume that entities correspond to well-defined and comparable concepts.

In cheminformatics and regulatory science, considerable effort has focused on handling substances without precise structural definitions, such as UVCBs. Methods like those of Kutsarova et al. [14] use generic structural encodings (e.g., G SMILES) and statistical sampling to generate representative constituent structures, reducing combinatorial complexity while preserving structural diversity.

These approaches operate at the level of individual substances and do not address broader semantic inconsistencies across regulatory datasets. Many entries still lack structure-resolvable identifiers or rely on heterogeneous definitions, limiting interoperability between data sources.

In the regulatory domain, previous work has highlighted challenges related to data fragmentation, heterogeneity of identifiers, and limited machine-readability of legal data [22]. In particular, chemical regulation introduces additional complexity due to the use of substance groups, mixtures, and analytical parameters that do not correspond to single chemical structures. Closer to the regulatory domain, Sepehri et al. [23] developed the TOXIN knowledge graph, which integrates toxicological data for EU cosmetic ingredient safety assessment using ontology-driven semantic technology, demonstrating that such approaches are applicable to regulatory chemical data. However, these approaches address single-domain integration and do not characterise the broader semantic mismatch across regulatory datasets. While prior studies have focused on improving data accessibility or applying FAIR principles to regulatory datasets, the fundamental

mismatch between chemical identity and regulatory substance definitions has not been systematically characterised. The present study addresses this gap.

3. Methods

Our analysis comprises the following steps:

Data Integration Substance records were compiled from a comprehensive set of 18 European regulatory and monitoring sources and one Flemish administrative list. These include ECHA obligation lists (Candidate List [24], Authorisation List [25], Restriction List [26], POPs List [27], EU Positive List [28], and the CLH Harmonised Classification and Labelling List [29]), ECHA activity lists (Restriction Process [30], SVHC Identification [31], Authorisation Process [32], Dossier Evaluation [33], CLH Process [34], Substance Evaluation [35], POPs Process [36], PBT Assessment [37], and ED Assessment [38]), as well as REACH registration data [39]. Additional sources comprise the EU Pesticides Database (Active Substances) [40], the OSPAR List of Chemicals for Priority Action [41], and a Flemish administrative substance list developed in the context of water quality modelling [42]. This Flemish *summation substances* list reflects the heterogeneity of *substance names* in Flemish environmental permits and monitoring. For each entry, available identifiers (CAS number, EC number, and substance name) were collected.

Where possible, substances were linked to structure-based identifiers (InChIKey) via PubChem [9]. The resulting dataset establishes a unified mapping between regulatory entries, their identifiers, and their originating source lists.

Chemical Identity and Linkability Each substance entry was classified into one of several types reflecting its degree of chemical specificity: structure-defined molecule, substance group, mixture/UVCB, analytical parameter, or administrative entry.

Based on this classification, a seven-tier linkability taxonomy was defined, ranging from tier 1 (directly structure-resolvable via InChIKey) through partially resolvable entries (tiers 2–3), unresolved CAS numbers (tier 4), mixtures and UVCBs (tier 5), and administrative entries (tier 6), to tier 7 (no usable identifier). This taxonomy expresses the degree to which regulatory substances can be unambiguously linked across datasets.

Identifier Consistency Analysis For entries with both CAS numbers and InChIKeys, bidirectional mappings were analysed to assess identifier consistency. This analysis evaluates whether CAS numbers uniquely map to structures and vice versa.

Structure Coverage and Cross-List Overlap For each regulatory source, the proportion of structure-defined substances was calculated. Overlap between sources was analysed using structure-based matching (i.e., using InChIKey) and set similarity measures.

Analysis of Non-Structure Substances Substances without structure identifiers were analysed using text-based methods, including embedding-based clustering [43,44,45] and similarity matching to chemical taxonomies (i.e., using ChemOnt), to assess whether semantic techniques can recover meaningful structure or classification [46,20]. Each substance name and ChemOnt class label was encoded using the `all-MiniLM-L6-v2` sentence-transformer model [21,47,48], producing a 384-dimensional dense embedding vector per substance entry.

Embedding Model Validation Using ChemOnt Ground Truth To evaluate the ability of text-based embeddings to recover chemical classifications from substance names, an internal validation was performed using structure-defined substances as ground truth. Substances with an unambiguous InChIKey were linked to the ChemOnt taxonomy through ClassyFire classification [20], providing reference assignments at the levels of direct parent, subclass, class, superclass, and kingdom.

Substance names and ChemOnt class labels were encoded using pre-trained transformer models (`all-MiniLM-L6-v2`, `all-mpnet-base-v2` [49], `allenai/scibert_scivocab_uncased` [50], and `dmis-lab/biobert-v1.1` [51]) within the *sentence-transformers* framework [21,47,48]. Cosine similarity was used to rank all ChemOnt classes for each substance name.

Performance was assessed using standard information retrieval metrics, including Hit@k ($k = 1, 3, 5, 10$) and Mean Reciprocal Rank (MRR) [46]. To account for the hierarchical structure of ChemOnt, Hit@1 was additionally computed for each taxonomy level. A weighted lenient MRR was introduced to summarise performance across levels, assigning ordinal weights of 1.00 (direct parent), 0.75 (subclass), 0.50 (class), 0.25 (superclass), and 0.10 (kingdom), thereby favouring more specific classifications.

This validation provides a principled comparison of embedding models and demonstrates the extent to which semantic representations of substance names approximate structure-based chemical taxonomy assignments.

Knowledge Graph Construction All data were integrated into an RDF knowledge graph using SKOS [52] for taxonomy representation. Structure-defined substances were linked to ChemOnt classes and to ChEBI entities via InChIKey, enabling integration of biological hazard annotations [53].

Workload Estimation For non-structure substances, interoperability requires manual assessment of pairwise relationships (equivalence, subset, overlap, disjoint). The required effort scales quadratically with the number of such entries as $\binom{N}{2} = N(N-1)/2$, where N is the number of non-structure entries under consideration.

Priority Scoring To illustrate the analytical value of the integrated knowledge graph (C5), a composite priority score was computed by combining regulatory list membership with biological role annotations from ChEBI [10], following the ToxPi framework [54]. This score is illustrative only and does not constitute a formal risk assessment.

4. Results

Overview of the Integrated Knowledge Graph We constructed an integrated knowledge graph to analyse the representation and interoperability of regulatory substances across datasets. Figure 3 provides an overview of the resulting graph structure and its main semantic layers. The graph is organised into four layers: an input layer of 18 regulatory sources; a core layer of ∼17,500 regulatory substance entities represented as SKOS concepts; linking layers connecting substances to structure-based identifiers (InChIKey), chemical classifications (ChemOnt), and biological roles (ChEBI); and derived layers comprising linkability tiers, embedding-based semantic clusters, and annotations (∼46,600 links).

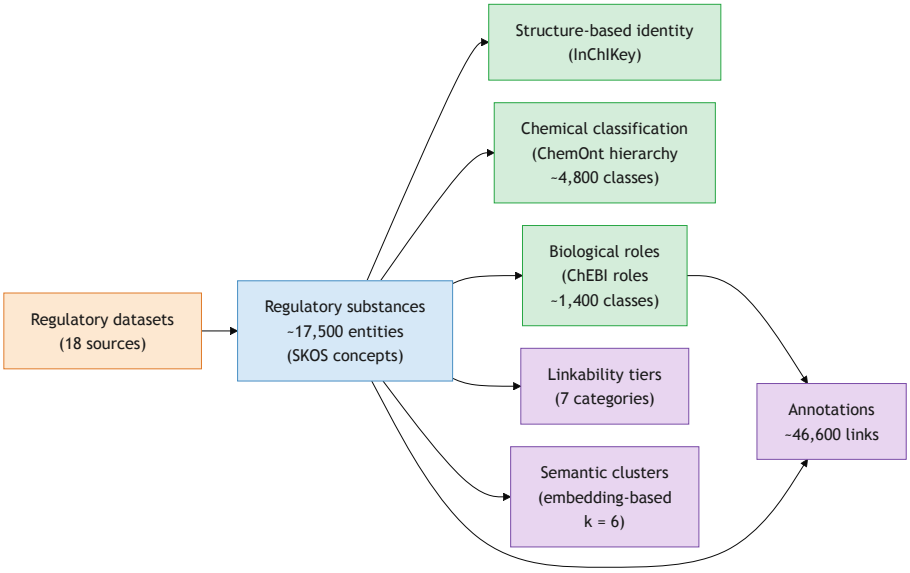

Figure 3. Overview of the integrated knowledge graph constructed from 18 regulatory datasets. The graph flows from the input layer of regulatory sources through the core layer of regulatory substances (SKOS concepts) to linking layers (InChIKey, ChemOnt, ChEBI) and derived layers comprising linkability tiers, semantic clusters, and annotations.

Dataset Overview (C1) Integration of 18 regulatory sources yielded 41,813 records, of which 63% could be linked to a defined molecular structure via InChIKey.

Heterogeneity of Substance Definitions A significant share of regulatory entries (37%) does not correspond to a defined chemical structure, consistent with Kutsarova et al. [14]. These include mixtures (UVCBs), substance groups, analytical parameters, or entries without usable identifiers, reflecting differences in how "substances" are defined across regulatory contexts.

Figure 4A shows that heterogeneity varies widely between regulatory lists: some sources consist almost entirely of structure-defined molecules, whilst others contain few structure-defined entries.

Applying the seven-tier linkability taxonomy (Section 3; Figure 4B) shows that 62.5% of entries (20,909) are directly linkable via InChIKey (tier 1). Only a small fraction consists of partially resolvable groups (tiers 2–3; $<0.2\%$).

The remaining 37.5% cannot be directly linked to a structure: 11.6% have unresolved CAS numbers (tier 4), 10.8% are mixtures or UVCBs (tier 5), $<0.1\%$ are administrative entries (tier 6), and 14.9% lack any usable identifier (tier 7).

More than one-third of regulatory entries cannot be mapped to a unique chemical structure, and even CAS numbers do not guarantee unambiguous identification. Consequently, structure-based integration is limited to part of the data, and the non-structure fraction requires different analytical treatment.

Limitations of CAS Numbers (C2) CAS numbers are not reliable unique identifiers. Bidirectional mapping analysis reveals that 195 InChIKeys are associated with more than one CAS number (up to 7 per structure), while 4 CAS numbers resolve to two distinct

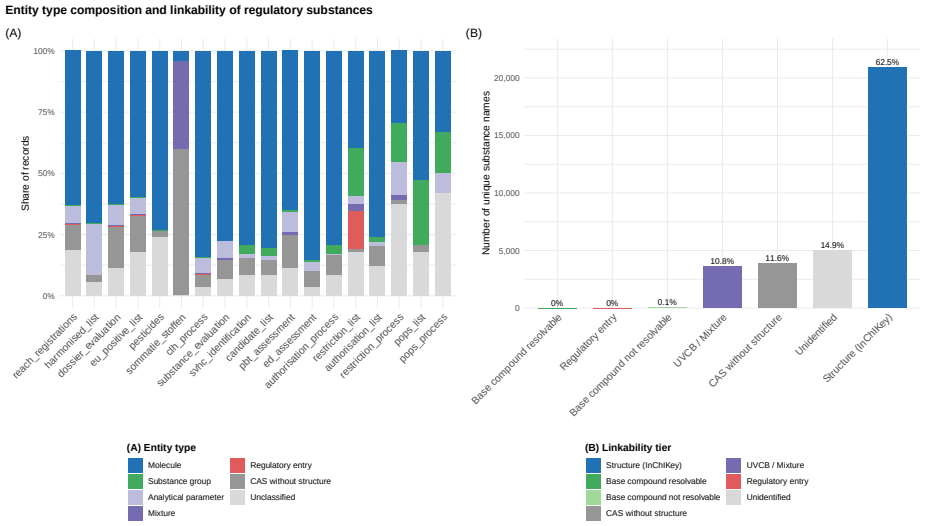

Figure 4. (A) Entity type composition across 18 regulatory sources. Each bar shows the proportional breakdown of substance entries into seven mutually exclusive types (Molecule, Substance group, Analytical parameter, Mixture, Regulatory entry, CAS without structure, Unclassified), illustrating that the mix of entity types differs substantially between regulatory instruments. (B) Linkability taxonomy applied to all 33,452 unique regulatory substance entries. A total of 20,909 entries (62.5%) can be directly linked to a chemical structure via InChIKey (tier 1), while 4,981 entries (14.9%) lack any usable identifier. Together, the panels show that structural linkability is unevenly distributed across sources and that more than one-third of all entries cannot be resolved to a unique chemical structure.

InChIKeys. These inconsistencies arise from differences in stereochemistry, salt forms, and context-dependent registration practices [8,55]. CAS-based cross-dataset linking therefore introduces ambiguity rather than resolving it, making CAS numbers insufficient as a backbone for interoperability.

Limited Interoperability Across Regulatory Lists Cross-list analysis of the 17,523 structure-resolved InChIKeys shows that 70.0% (12,270 entries) appear in only a single regulatory source, while just 30.0% (5,253 entries) are shared across two or more lists. This limited overlap reflects the domain-specific design of regulatory instruments rather than a shared chemical reference framework.

Failure of Semantic Approaches for Non-Structure Substances (C3) Embedding-based clustering of non-structure substance names yields semantically coherent groups but does not resolve chemical identity. While clusters reflect lexical or contextual similarity among substance descriptions, they do not correspond to well-defined molecular entities (Figure 5A), confirming that neither lexical context nor pre-trained language models can recover chemical structure from regulatory substance names.

To evaluate whether semantic similarity can recover chemical meaning, we attempted to map non-structure substances to ChemOnt taxonomy classes using cosine similarity between sentence embeddings of substance names and ChemOnt class labels. Four models were evaluated: `all-MiniLM-L6-v2` (miniLM), `all-mpnet-base-v2` (mpnet), `allenai/scibert\_scivocab\_uncased` (SciBERT), and `dmis-lab/biobert-v1.1` (BioBERT). The distribution of top-1 similarity scores is heavily skewed

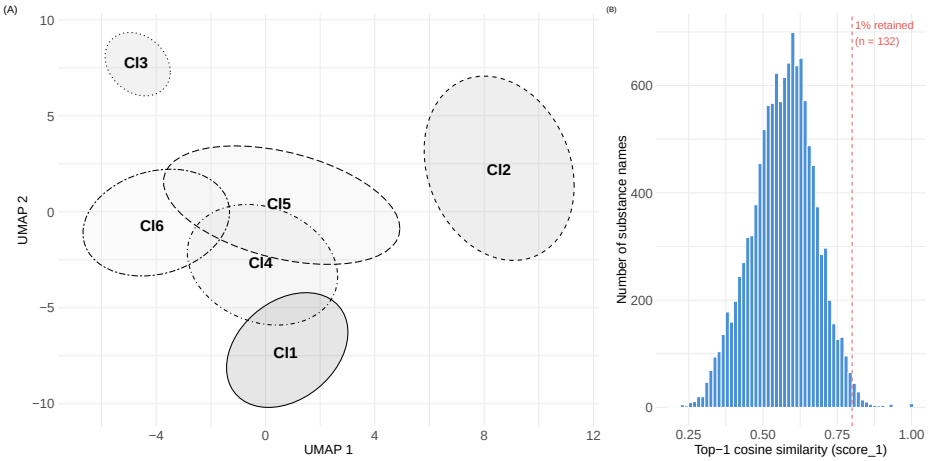

Figure 5. (A) UMAP projection of sentence embeddings for 12,499 non-structure regulatory substance entries, grouped by k-means cluster (k = 6). Each point represents a unique substance name; spatial proximity reflects semantic similarity in the 384-dimensional embedding space. Clusters: Cl1 = Reaction masses; Cl2 = Complex substances (UVCB with structure); Cl3 = Petroleum & coal tar fractions; Cl4 = Inorganic compounds & metal salts; Cl5 = Trade names, codes & biological materials; Cl6 = Polymers, fatty acids & surfactants. (B) Distribution of top-1 cosine similarity scores [46] between substance name embeddings and ChemOnt class label embeddings across all 12,499 candidate substance-class pairs, shown here for `all-MiniLM-L6-v2`. The dashed line indicates the applied threshold ($s_1 > 0.80$); 132 entries (1.1%) are retained as matches.

towards lower values (Figure 5B). Cosine similarity scores are not comparable across models: scores for the 200th-ranked match range from approximately 0.80 for miniLM to 0.94 for BioBERT, reflecting differences in embedding geometry rather than differences in match quality. A fixed threshold such as $s_1 > 0.80$ is therefore model-dependent and arbitrary; it was applied here for the purpose of bounding a manual evaluation, not as a principled quality criterion.

Manual inspection of the top-200 matches per model revealed that the majority of retained assignments are chemically incorrect, regardless of model. Correct assignments were largely limited to cases where the substance name closely resembles the ChemOnt class label (e.g. *Phenols* → *Phenols*, *Alkyl mercury compounds* → *Alkyl mercury compounds*), indicating that high similarity scores reflect surface-form overlap rather than semantic understanding. Systematic errors were present across all models: miniLM conflates organic and inorganic species and confuses elements sharing a common prefix (e.g. *organostannic compounds* → *organoarsenic compounds*); mpnet consistently misassigns all nickel sulfide variants to *nitrile sulfides*, a token-level artefact of the chemical symbol Ni being read as the start of "nitrile"; SciBERT and BioBERT, pre-trained on biomedical literature, exhibit severe score inflation (a score of 0.93 on an incorrect match provides less information than a score of 0.85 on a correct one) and produce repeated false anchors to specific heterocyclic and lipid ChemOnt classes for structurally unrelated substances. Trade names and reaction products were assigned arbitrary classes with no verifiable basis across all models.

Score magnitude is not a reliable indicator of match correctness. Models with the highest score floors (SciBERT, BioBERT) performed worst on both manual inspection and quantitative validation, demonstrating that inflated cosine scores reflect poor cali-

bration rather than genuine chemical understanding. Embedding-based methods cannot compensate for the absence of structure-based identifiers, and fail systematically for substance groups, mixtures, and analytically defined entries.

To validate this failure quantitatively, the four models were evaluated against a structure-derived ChemOnt ground truth comprising 28,204 ClassyFire-assigned substances. These results, shown in Table 1, indicate that `all-MiniLM-L6-v2` achieved the highest Hit@1 of 5.75% and MRR of 0.093 — even this best result implies the correct class is retrieved in fewer than 1 in 17 attempts. All other models scored below Hit@1 = 1%. Retrieval performance decreased further with increasing taxonomic abstraction: Hit@1 fell to 3.25% at the subclass level and below 1% for class, superclass, and kingdom. A hierarchy-aware weighted MRR of 0.099 was only marginally higher than the strict MRR, confirming that partial matches at higher levels of the ChemOnt hierarchy do not rescue overall classification quality.

Table 1. Performance of embedding models for ChemOnt classification.

Model	Hit@1	Hit@3	Hit@5	Hit@10	MRR
all-MiniLM-L6-v2	0.0575	0.0986	0.1234	0.1621	0.093
all-mpnet-base-v2	0.0093	0.0152	0.0184	0.0247	0.015
SciBERT	0.0040	0.0071	0.0102	0.0131	0.008
BioBERT	0.0047	0.0077	0.0099	0.0150	0.009

Scale of the Interoperability Problem (C4) For the more than 11,000 non-structure entries, full manual reconciliation would require on the order of hundreds of millions of pairwise comparisons. Under conservative assumptions, this corresponds to several thousand person-years of expert effort.

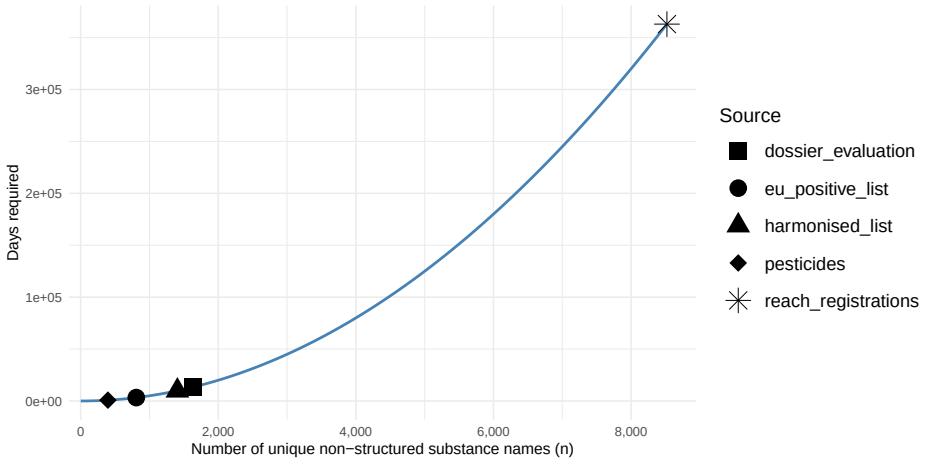

Figure 6. Estimated pairwise curation workload for non-structure entries across the combined European regulatory dataset, broken down by source. Reach registrations correspond to approximately 360,000 person–days (1,800 person-years) under an assumed throughput of 100 pair assessments per person-day. The quadratic growth of the workload illustrates that manual reconciliation is not scalable, reinforcing the need for structure-based and ontology-driven representations at the source.

Figure 6 quantifies this workload across all regulatory sources, illustrating both the absolute scale and the quadratic growth that makes each additional non-structure entry disproportionately costly. Even moderate increases in the number of unresolved entries lead to a rapidly escalating curation burden.

Manual pairwise reconciliation at this scale is not operationally feasible. This result highlights a fundamental constraint: interoperability cannot be achieved through retrospective harmonisation of heterogeneous substance definitions.

Instead, the only scalable strategy is to address the problem at its source. By requiring structure-resolvable identifiers and ontology-based representations in regulatory definitions wherever chemically applicable, the number of entries requiring manual reconciliation can be structurally reduced rather than managed post hoc.

Knowledge Graph Integration and Priority Scoring (C5) Structure-defined substances can be successfully integrated across regulatory lists, chemical taxonomies (ChemOnt), and biological annotations (ChEBI) using InChIKey as a linking key. This enables cross-domain queries that are infeasible on fragmented regulatory datasets: for example, identifying substances that appear on three or more ECHA lists and carry a documented biological role in ChEBI, or retrieving all members of a given ChemOnt class that are subject to authorisation requirements. As an illustration of this analytical potential, a composite priority score combining regulatory list membership with ChEBI biological role annotations was computed following the ToxPi framework [54]. Figure 7 plots regulatory versus biological scores for all scored substances; 2,2-bis(4-hydroxyphenyl)propane and dibutyl phthalate exemplify substances with both broad regulatory presence and multiple documented biological roles. This score is illustrative and does not constitute a formal risk assessment.

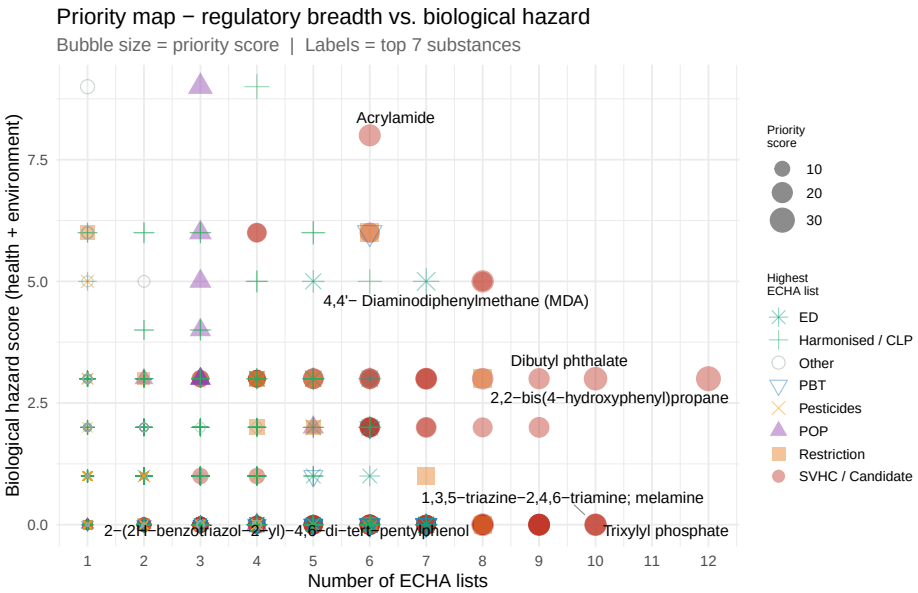

Figure 7. Bubble chart of composite priority space: regulatory list score versus biological hazard score for all scored substances. Bubble size is proportional to the number of regulatory source memberships. Substances in the upper-right quadrant combine broad regulatory coverage with documented biological hazard evidence.

5. Implications

Structural Linkability and Its Limits (RQ1) The results demonstrate that interoperability of regulatory substance data is fundamentally constrained by the absence of a shared and formalised notion of chemical identity. While structure-based identifiers such as InChIKey enable unambiguous linking across datasets, a substantial fraction of regulatory entries cannot be expressed in this form. Instead, regulatory "substances" frequently represent heterogeneous entities, including mixtures, substance groups, and analytical parameters.

This heterogeneity is not merely a data quality issue but reflects a structural mismatch between chemical reality and regulatory abstraction. As a consequence, semantic relationships between regulatory entries, such as equivalence, subset relations or partial overlap, remain implicit and undefined. Without explicit formalisation of these relationships, automated integration across regulatory datasets is inherently limited.

Unreliability of CAS-based Identification (RQ2) The analysis confirms that commonly used identifiers in regulatory contexts, particularly CAS numbers, do not provide a reliable foundation for interoperability. Although widely adopted, they exhibit both one-to-many and many-to-one mappings with chemical structures, reflecting differences in stereochemistry, salt forms, and context-dependent definitions.

This finding challenges the widespread assumption in regulatory data integration that such identifiers can serve as globally unique references. In practice, relying on them introduces ambiguity rather than resolving it, undermining their suitability as a primary reference mechanism for cross-dataset linking.

Failure of Semantic Compensation (RQ3) This study shows that modern embedding-based methods cannot compensate for the absence of structure-based identifiers in regulatory substance data. While they can group substance names into loosely coherent semantic clusters, they fail to recover true chemical identity or reliably map entries to ChemOnt classes, with only a negligible number of correct assignments.

These limitations are inherent to the data rather than the models: textual descriptions of mixtures, groups, and analytical entries lack the information required to infer molecular structure. As a result, semantic similarity largely reflects surface-level lexical overlap rather than genuine chemical relatedness.

Sentence embeddings cannot therefore substitute for structure-resolvable identifiers; reliable chemical classification and interoperability thus require explicit structural representation, such as InChIKeys, at the source.

The Combinatorial Barrier to Retrospective Harmonisation The scale of the interoperability problem becomes evident when considering the effort required for manual reconciliation. For non-structure substances, establishing semantic relationships requires pairwise comparison, leading to quadratic growth in workload. The estimated effort of several thousand person-years demonstrates that retrospective correction is not a viable strategy at scale, interoperability must be built into regulatory data models by design.

Towards Ontology-Driven Regulatory Data Infrastructures In contrast to regulatory datasets, cheminformatics infrastructures such as PubChem and ChEBI demonstrate that large-scale interoperability is achievable when chemical identity is explicitly represented and linked to formal ontologies. Structure-based identifiers, combined with hierarchical classifications and machine-readable knowledge graphs, enable consistent and scalable integration across scientific and regulatory domains.

The priority scoring example illustrates that once chemical identity is formalised and linked across datasets, higher-level aggregations combining regulatory and biological information become straightforward to compute.

The results of this study show that similar principles can be applied to regulatory data. Structure-defined substances can be successfully linked across datasets and enriched with chemical and biological annotations via InChIKey and formal ontology-based classifications.

Implications for FAIR Chemical Data in Europe The findings have direct implications for the implementation of FAIR (Findable, Accessible, Interoperable, Reusable) principles in European chemical regulation. Current regulatory datasets fall short of interoperability primarily because they lack globally unique, persistent, and machine-readable identifiers grounded in chemical structure.

Achieving FAIR chemical substance data requires a shift from document-centric and identifier-fragmented approaches towards linked data infrastructures [56,57] in which substances are represented as resolvable URIs, grounded in structure-based identifiers and connected through formal ontologies. This shift is not purely technical: it requires alignment between data modelling practices and the legal definitions of substances in regulatory frameworks.

Limitations and Scope This study focuses on datasets derived from ECHA and related European regulatory sources, and on the use of InChIKey as the primary structure-based identifier. While the results are consistent across the analysed datasets, further work is

needed to assess the generality of these findings across other regulatory domains, jurisdictions, and substance categories.

Furthermore, not all regulatory concepts can or should be reduced to individual molecular structures. Mixtures, UVCBs, and analytical parameters are legitimate and necessary constructs in regulation. However, their current representation lacks formal semantics, limiting their integration with structure-based systems. Future work should explore hybrid models that explicitly represent these entities as sets, classes, or functions over well-defined chemical structures.

From Data Integration to Regulatory Design The interoperability problem is not primarily a technical integration challenge, but a consequence of how substances are defined and represented in regulation. Efforts to improve interoperability must therefore extend beyond data engineering to include the principled design of regulatory concepts, vocabularies, and data standards.

Embedding structure-based identifiers, ontology references, and machine-readable semantics directly into regulatory processes [58,18] would significantly reduce ambiguity and enable scalable integration. Without such changes, the gap between regulatory data and scientific data infrastructures will continue to widen, limiting the effectiveness of data-driven environmental governance.

In practice, this implies that regulatory data infrastructures should: (i) include structure-based identifiers (e.g. InChIKey) wherever applicable; (ii) represent substance groups and mixtures using explicit semantic models (e.g. set- or class-based representations); (iii) publish substance data as linked data with persistent URIs.

6. Conclusion

In this study, we show that interoperability of chemical substance data in European regulatory frameworks is fundamentally limited by the lack of a shared and formalised notion of chemical identity. Analysis of 18 regulatory datasets reveals that a substantial fraction of substances cannot be unambiguously linked to a molecular structure, and that commonly used identifiers such as CAS numbers are insufficient for reliable and consistent cross-dataset integration.

The three research questions are answered as follows. RQ1: A seven-tier linkability taxonomy (Section 3) applied to 41,813 records shows that 62.5% of regulatory entries (20,909) resolve directly via InChIKey (tier 1), while the remaining 37.5% cannot be linked to a defined chemical structure: 11.6% carry unresolvable CAS numbers (tier 4), 10.8% are mixtures or UVCBs (tier 5), and 14.9% lack any usable identifier (tier 7). RQ2: CAS numbers prove unreliable as unique identifiers; bidirectional mapping shows that 195 InChIKeys are each linked to more than one CAS number (some carrying up to seven) and 4 CAS numbers resolve to two distinct InChIKeys, with inconsistencies arising from differences in stereochemistry, salt forms, and registration context. Cross-list analysis further shows that 70% of structure-resolved InChIKeys appear in only a single regulatory source, reflecting domain-specific rather than shared chemical reference frameworks. RQ3: Semantic and embedding-based approaches cannot compensate for the absence of structure-based identifiers; the best-performing model achieves a ChemOnt Hit@1 of only 5.75%, meaning the correct chemical class is retrieved in fewer than 1 in 17 attempts, and all other models scored below Hit@1 = 1%. Manual

inspection reveals that high cosine similarity scores reflect surface-form lexical overlap rather than chemical understanding, a failure inherent to the data: textual descriptions of mixtures, substance groups, and analytical parameters simply do not contain sufficient information to infer molecular structure.

These limitations directly affect semantic interoperability. Many regulatory definitions lack explicit structure-based identifiers and formal semantics, leaving relationships such as equivalence, inclusion, or overlap implicit. This is particularly evident for substance groups and analytical parameters, which often represent sets or transformations of underlying chemical entities (Section 4). Without explicit formalisation of these relationships, interoperability and automated reasoning remain severely constrained.

The scale of the problem becomes clear when considering the effort required for manual reconciliation. For the more than 11,000 non-structure entries, establishing semantic relationships requires pairwise comparison, leading to quadratic growth in curation workload. Under conservative assumptions this corresponds to approximately 630,000 person-days (3,150 person-years) of expert effort, demonstrating that retrospective harmonisation is not operationally feasible. In contrast, structure-defined substances can be successfully integrated at scale: the knowledge graph constructed in this study (Section 4) links regulatory entities to InChIKey, ChemOnt, and ChEBI annotations across approximately 46,600 cross-domain links, enabling cross-domain queries and composite priority scoring (Section 3) that are infeasible on fragmented regulatory datasets.

Our results indicate that interoperability cannot be achieved through post hoc technical solutions alone. Instead, it requires changes at the level of data modelling and regulation: the inclusion of structure-based identifiers (e.g. InChIKeys), explicit formal semantic models for substance groups and mixtures, and publication of regulatory data as linked data with persistent, resolvable URIs. Without such changes, fragmentation between regulatory and scientific data infrastructures will persist.

Future work should address several open directions. First, formal semantic representations are needed for mixtures, UVCBs, and substance groups that can express their compositional and set-theoretic structure in a machine-readable way, for example representing a substance group as an explicit set of member InChIKeys, or an analytical sum parameter as a function over named constituents. Second, these representations must be translated into operational standards for regulatory data exchange, embedding interoperability requirements directly into the design of future regulatory instruments and reporting obligations. Third, the generality of the findings should be assessed beyond European ECHA datasets, across other regulatory domains, jurisdictions (e.g. US EPA and UK HSE), and substance categories where similar semantic mismatches are likely present. Fourth, the 37.5% of non-structure entries that currently resist automated linking remain an open challenge: curated resolution workflows, possibly guided by the linkability taxonomy introduced here, could provide a principled starting point for targeted harmonisation. Finally, the knowledge graph infrastructure developed as part of this study provides a foundation for further work on automated reasoning, regulatory impact assessment, and cross-domain prioritisation, capabilities that depend critically on the continued expansion of structure-grounded and ontology-linked regulatory data.

Data Availability

All data, code, and materials supporting this study are publicly available. The complete dataset, including the integrated knowledge graph, analysis scripts, and figures, is archived on Zenodo at `https://doi.org/10.5281/zenodo.xxxxxxx`[1].

Supplementary Materials

The integrated knowledge graph (RDF/Turtle), processed datasets from 18 regulatory sources, analysis scripts, and all figures are available via the Zenodo archive associated with this publication.

Declaration on the Use of Generative AI

During the preparation of this work, the author used Claude Code (Anthropic, model: `claude-sonnet-4-6`) for the following purposes: (1) writing, debugging and documenting analysis code in R, Bash, and Python; and (2) assisting with the system configuration to enable local execution of transformer-based embedding models. After using this tool, the author reviewed and edited all content as needed and takes full responsibility for the publication's content.

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